

Naukri AI Auto-Apply Agent

Complete Start-to-Finish System Algorithm Walkthrough

Step 1: Initialization & Resume Parsing

- **Load Configuration:** The agent reads settings from config.yaml (credentials, search terms, exclusions, daily caps).
- **Parse Resume Profile:** The ResumeParser reads your PDF resume using PyMuPDF or Tesseract OCR, extracting skills, work experience, and job history into a structured ResumeProfile object.

Step 2: Job Scraping Phase (Inside search_all)

- **Build Search URLs:** For each Keyword and Location combination, the bot constructs target search URLs with experience, salary, freshness, and sorting query parameters.
- **Scrape Job Cards:** Playwright loads pages sequentially (up to max_pages) and parses search result job cards (Title, Company, URL, Job ID, Location, Experience, Basic Skills).
- **Deduplicate in Dictionary:** Jobs are added to a dictionary using their Job ID as the key (all_jobs[job_id] = job). This prevents duplicate processing of the same listing.
- **Convert to List:** The dictionary values are converted to a standard list (list(all_jobs.values())) and returned to the agent loop.

Step 3: Prioritization Phase (Inside _process_jobs)

- **Calculate Heuristic Score:** For each job in the list:
 - If enable_heuristics is true: Computes TF-IDF vector similarity between your resume text and the job card. Applies boosts for keyword title matches (+0.15), top resume skills (+0.10), and posted freshness (+0.10 or +0.05).
 - If enable_heuristics is false: Score defaults to 0.0.
- **Build Priority Queue:** Jobs are pushed into a Max-Heap priority queue:

```
heapq.heappush(job_queue, (-score, idx, job))
```

Step 4: Loop & Pre-Filtering Phase (For Each Popped Job)

- **Retrieve Job:** The bot pulls the highest priority job using heapq.heappop(job_queue).
- **Daily Limit Check:** Verifies if the daily application cap has been reached.
- **Database Deduplication:** Checks SQLite DB to skip jobs already applied to.
- **Scam Filter Check (Pass 1):** The ConsultancyScamSpecification scoring engine scans the Title and Company Name for blocklisted agencies, BPO terms, or recruiter patterns. If the suspicion score hits 100+, the job is skipped immediately.

Step 5: Detail Scraping & Re-Filtering

- **Load Detail Page:** Playwright navigates the browser to the job's unique detail URL.
- **Extract Full Text:** Scrapes the complete job description, detailed requirements, and key skills.
- **Scam Filter Check (Pass 2):** Scans the full description text for hidden scam indicators (e.g., training fees, registration fees, laptop charges). If the score hits 100+, the job is skipped.
- **Second Similarity Filter:** Computes TF-IDF similarity on the full description text. Jobs scoring below 0.04 are skipped.

Step 6: AI Evaluation & Application

- **AI Matching (Gemini):**
 - *If `enable_matching` is true:* Sends your resume profile and the full job description to Gemini AI to generate a match score, missing/matching skills, screening answers, and a final `should_apply` decision.
 - *If `enable_matching` is false:* Bypasses matching and assumes `should_apply` = True.
- **Dry Run Check:** If `dry_run`: true, updates database status but skips browser submission.
- **Apply & Submit:** The browser applier fills out custom form answers via Gemini, attaches the resume, and clicks submit.
- **Save Status:** Logs the final status in the database (APPLIED, FAILED, or SKIPPED).